Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/22/2006
PI Name:	Homsy, George Ph.D.		
Project Title:	MIXING IN MICROGRAVITY MANIPULATING TANGENTIAL STRESSES		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	93106-5070	**Congressional District:**	23
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	10/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04GA42G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This task is concerned with experimental, theoretical and computational studies of mixing in fluids under conditions that model the microgravity environment. Task Objective: 1. Study mixing due to time-dependent body forces (g-jitter) by computational methods for both harmonic and stochastic g(t). Characterize the mechanism and the degree of mixing as functions of the parameters. 2. Study mixing in drops due to the time- modulation of electrohydrodynamic stresses by computational methods. 3. Study mixing and shape distortion in drops by experimental methods 4. Study the enhanced transport of heat or mass in droplets being mixed by flow fields that result in chaotic advection. The mixing of fluids can be crucial in fuel utilization, combustion, crystal growth, and other applications in microgravity. There is little understanding of the role of transient accelerations on fluid mixing when density differences are present and how these mixing mechanisms depend on the magnitude, direction, and spectral content of g(t). . There is also little understanding of the extent of mixing and the enhancement		

	of heat or mass transfer in chaotically stirred fluids.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	All our studies of are of general interest and applicability to Earth scale processes that involve either very small scale systems or neutrally buoyant fluid pairs. These processes include electrospraying, emulsion technology, foams, and the migration of drops and bubbles.
Task Progress:	We have been active in three different areas: Mixing due to g-jitter. A PhD student, Vikram Siddavaram, and the PI have presented papers at the APS Division of Fluid Dynamics meeting in Seattle in 2004 and in Chicago in 2005. They have submitted a paper to the J. Fluid Mechanics that was favorably reviewed and, after revision, is now in press. This work treated the mixing in time dependent, harmonic body forces and characterized the transition from laminar to transiently turbulent flows, identified the underlying mechanisms as Kelvin-Hemholz and Rayeigh-Taylor instabilities, and found the criteria for transition between these different regimes. Mixing due to electrohydrodynamic stresses: One PhD student, Thomas Ward, has completed his degree and is now at UCLA. His most recent work dealt with steady, three dimensional chaotic streamlines within drops. Both theory and experiment were conducted. The theory identified certain key dynamical events in which particles jump from one nearly steady streamline to another. These key events were observed in some complicated painstaking experiments, with general agreement between them and the theory. This work has been presented at APS/DFD meetings and is in press in J. Fluid Mech. A second PhD student, Xiu Mei Xu, has studied drop deformation electric fields in the presence of a translation velocity, and has done both theory and experiment on this problem, with good qualitative agreement between the two. This work has been reported at the Chicago APS/DFD meeting in 2005 and is in press in J. Fluid Mechanics. Enhancement of heat or mass transfer by chaotic mixing: With a senior colleague, Prof. Christo Christov, the PI has studied the enhancement in transport rates due to chaotically stirred drops. The results show a remarkable resonance phenomena, with significant enhancements at particular frequencies, and the discovery of underlying spatial structures, so called "strange eigenmodes" that explain the occurence of resonances.
Bibliography Type:	Description: (Last Updated: 03/22/2006)
Articles in Peer-reviewed Journals	Ward, T.; Homsy, G. M. "Chaotic streamlines in a translating drop with a uniform electric field " J. Fluid Mechanics , Jan-2006
Articles in Peer-reviewed Journals	Xu, Xiu-Mei; Homsy, G. M. "The settling velocity and shape deformation of drops in a uniform electric field" J. Fluid Mechanics , Jun-2006
Articles in Peer-reviewed Journals	Christov, C.; Homsy, G. M. "Enhancement of Transport from Drops by Modulated Electric Fields" Physics of Fluids , Jul-2006
Articles in Peer-reviewed Journals	Siddavaram, V.; Homsy, G. M. "The effects of gravity modulation on fluid mixing. Part 1: Harmonic modulation" J. Fluid Mechanics , Jul-2006